



Principles of Binaural Room Simulation

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(Received 12 December 1991; revised version received 17 January 1992;
accepted 2 March 1992)

ABSTRACT

Binaural room simulation is a method of allowing for listening into non-existing spaces by means of a computer model that takes all acoustical relevant parameters of the environment into account. It is based on a combination of traditional sound-field modelling techniques and binaural technology. An introduction into the underlying principles is given. Special attention is paid to the treatment of physical data, optimisation methods and aspects of auralisation and real-time performance.

1 INTRODUCTION

It is the aim of virtual-environment settings to put humans into situations in which they have modal perceptions which do not correspond to their real environment but to a different 'virtual' one. To reach such a goal, percepts have to be created artificially to set up the desired virtual environment. Since humans have percepts of different modality (visual, auditory, tactile, gustatory, olfactory and coordinative) the creation of virtual environment is a multimodal task. Yet, depending upon the specific application, a rank order of the different modalities in terms of task-specific relevance can be set up. Consequently, some modalities can eventually be treated with less authenticity or even be neglected.

The sense of hearing is of paramount importance for interindividual communication amongst humans. For this reason, amongst others,

applications of virtual environment which include interindividual communication tasks must provide for appropriate treatment of the auditory modality.

To this end, consider an everyday listening situation with a sound being emitted by a sound source somewhere in a space surrounded by walls. Owing to the presence of the walls, which represent sound-reflecting surfaces, a complex sound field arises. It is not far from the mark to state that the physical properties of the acoustic environment are encoded in the sound field. A sound receiver who samples the sound field at a certain position will thus catch information not only about the emitted signal and the sound source, but also about the acoustic environment.

In the case of a human listener, the incoming sound is transformed by the external ear (pinna, head and torso) into sound pressure signals at the two ear drums. During that transformation process, spatial cues from the sound field are converted into temporal and spectral cues in the eardrum signals. The auditory system, consequently, can decode these cues and use the information rendered to form auditory percepts accordingly. The influence of the acoustic environment is mainly reflected in spatial attributes of the auditory percepts, such as direction, distance, spatial impression and extension.

The process of coding parameters of the acoustical environment into sound-field characteristics can be modelled on the computer, using a methodology called *room simulation*. Likewise, the transformation of sound-field cues into cues in the input signals to the ear drums can be modelled by a methodology called *binaural technology*. Both simulation methodologies in combined form are referred to as *binaural room simulation*. It is the purpose of this article to provide an introduction to binaural room simulation and to discuss some of its specifics.

2 REPRESENTATION OF THE SOUND FIELD

Acoustic environments are specified by physical properties which, in turn, have an influence on the sound field in the environments. Such properties are the geometric and acoustic data of sound reflecting surfaces, and the positions, orientations and directional characteristics of both the sound source and the receiver.

It is the task of binaural room-simulation systems to create auditory percepts which correspond to an environment as defined by a given set of these parameters. The binaural room-simulation process can be conceived as being composed from two operational parts: sound-field modelling and auralisation. In the first part, a computer model of the sound field is created,

based on an appropriate set of representative physical data of the environment. In the second part the results will be made audible.

In a condition with reflecting surfaces, listeners receive sound signals directly coming in from the sound source. But in addition to this direct sound, due to the existence of reflecting surfaces, a variety of reflected sound signals will impinge upon the listener from different directions, with different arrival times and different spectral shapes. This situation can be described using the concept of so-called virtual sound sources (see, e.g. Ref. 1). There, the sound field inside reflecting surfaces is described in terms of a 'cloud' of virtual sound sources surrounding the receiver in the absence of reflecting surfaces. To this end, each virtual source is assigned a linear filter by which the direct sound is filtered to obtain the virtual-source signal. The filters model the influence of the propagation medium and of the reflecting surfaces plus the directivity of the primary source.

In Fig. 1 we depict the structure of a binaural room-simulation system following the above ideas.

In the sound-field-modelling stage, the spatial distribution of mirror sources is computed, based on the physical parameters of the environment. Then, in the auralisation stage, binaural impulse responses of the room are computed and, by convolution with pre-recorded anechoic signals and subsequent reproduction over headphones, listening into the modelled (virtual) environment becomes possible. As an example, Fig. 2 shows a simple computer model of a concert hall with some sound rays, Fig. 3 the computed distribution of virtual sound sources and Fig. 4 the corresponding binaural impulse response.

Due to limited dynamic and limited spatial and temporal resolution of the human auditory system, the spatial distribution of mirror sources in the model can be quantised and, consequently, be simplified to a reasonable

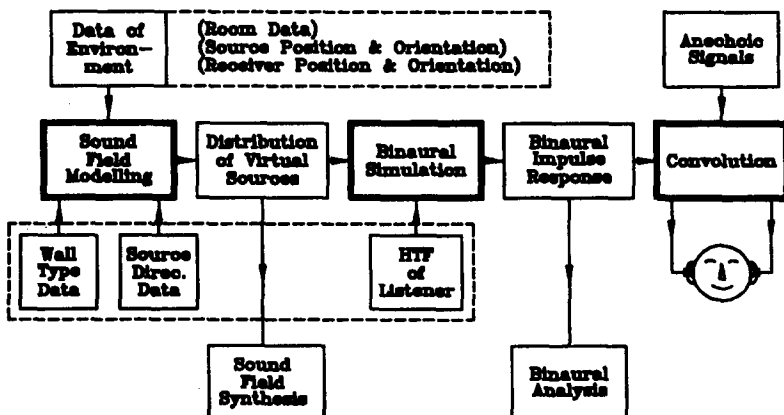


Fig. 1. Schematic of the structure of a binaural room-simulation system.

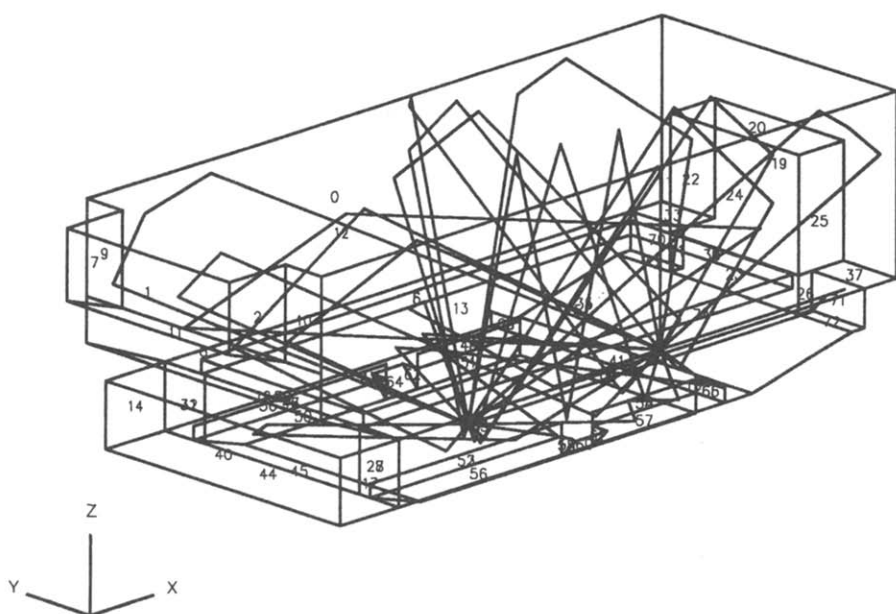


Fig. 2. Simple computer model of a concert hall. The surfaces are numbered. Some sound rays that start from the source and hit the receiver are shown.

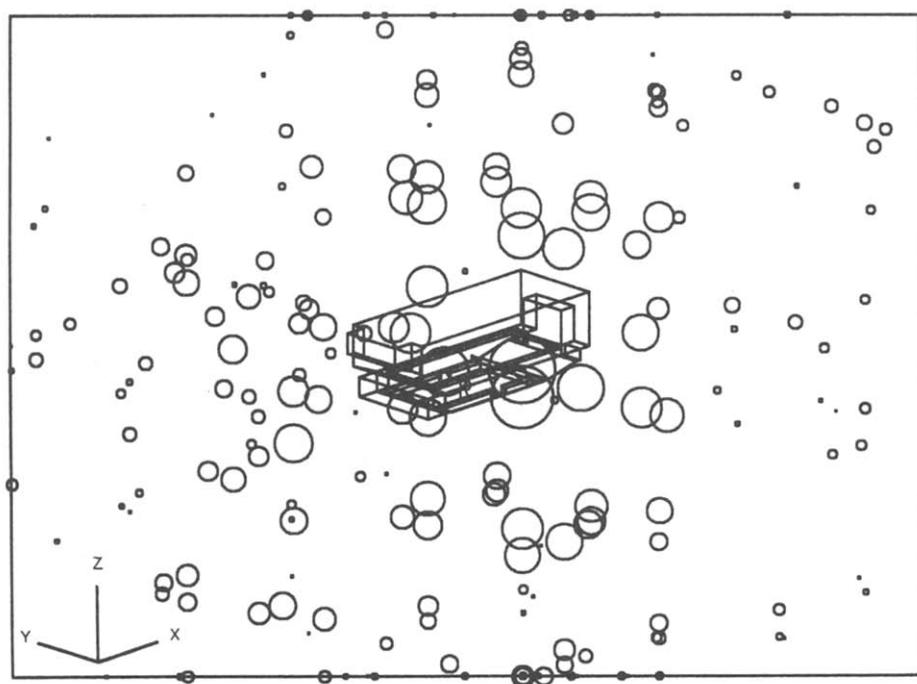


Fig. 3. Two-dimensional projection of a distribution of virtual sound sources as computed from the configuration as shown in Fig. 2. Each circle denotes a virtual sound source, the energy of which is decoded by the area of the circle.

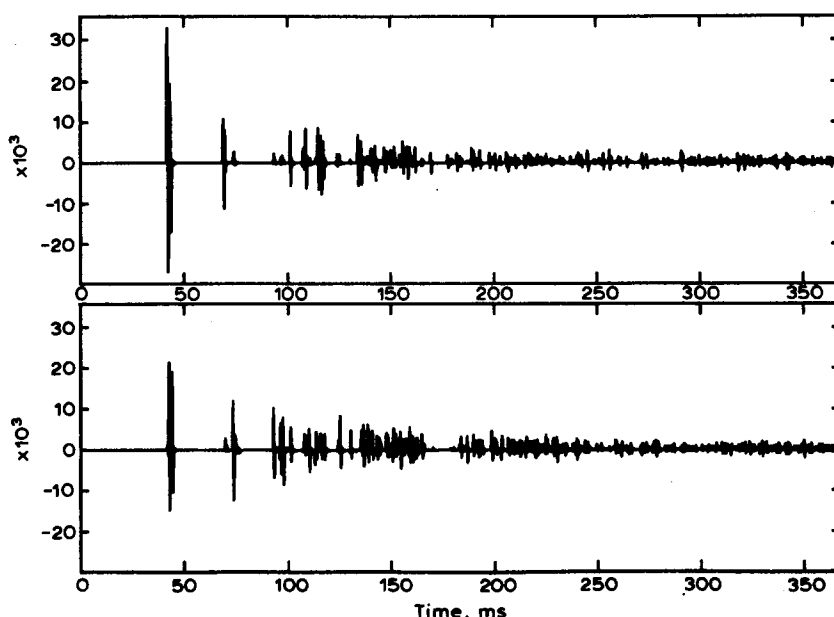


Fig. 4. First 370 ms of the binaural impulse response derived from the distribution of virtual sound sources as shown in Fig. 3. The upper half shows the impulse response at the left ear.

extent. This means, in consequence, that practically every sound field can be simulated by a finite number of virtual sound sources.

It should be noted here, that the description of a sound field by means of virtual sound sources is not restricted to models that are based on geometrical acoustics but can also be applied in different algorithms, e.g. numerical solutions of the wave equations. Also, measurement techniques suitable for obtaining distributions of virtual sources from existing spaces have been developed.^{2,3} Using the tools available today, comparison between virtual and real acoustic environments can be performed both objectively and perceptually.

3 SOUND-FIELD MODELLING BY MEANS OF GEOMETRICAL ACOUSTICS

First attempts to model sound fields as occurring in rooms of complex geometry have used scaled-down physical models of the rooms.⁴ In recent years, the scale-modelling technique has matured, and even binaural listening into such models is now possible as a routine technique, based on the use of advanced scale dummy-heads.^{5,6} However, due to the considerable efforts required to build scale models, they are mainly used to work on

room-acoustics problems of some commercial significance, such as concert-hall design. For the creation of auditory virtual environments, scale models are not generally available.

Recently, due to the tremendous progress in computer technology, sound-field modelling by computer has become a realistic alternative to physical scale modelling. In general terms, the computation of a sound field in a room is a straightforward linear problem which can be treated by solving Kirchhoff's equations for the complex boundary conditions as given by the surrounding surfaces. However, a general analytic solution to the problem cannot be found with reasonable effort since the formulation of the boundary conditions is extremely difficult. Numerical solutions can be obtained by using finite-element or boundary-element methods (e.g. Ref. 7). Recent progress comes from the use of time-iterative boundary-element methods,⁸ but due to the computational power needed for them, these methods are still restricted to comparatively small rooms or low frequencies.

Geometrical acoustics is based on a special solution of the wave equation which only holds when the wavelength of the sound is small compared to the geometrical dimensions of the surrounding surfaces, and large compared to their roughness and the curvature. If this condition is fulfilled, sound-pressure waves can be treated like light waves; for example, the geometrical law of reflection holds.

Valuable computer models have been developed based on geometrical acoustics. In general, the disadvantages of those models is that a-priori inclusion of wave-propagation effects like diffuse reflections or diffraction is not given. Particularly, diffuse reflections play an important role in room acoustics. However, as will be explained later, certain extensions to models based on geometrical acoustics can be applied in order to mimic diffuse reflections to a certain extent.

Two basically different algorithms are known for sound-field modelling using geometrical acoustics: the mirror-image method and the ray-tracing method. Both methods have been applied to room acoustics for quite some time^{9,10} and have more than once been compared with respect to their efficiency (e.g. see Refs 11–13).

In the following, we give an overview on different implementations and discuss their advantages. For a more detailed description of the algorithms we refer to the literature.

3.1 The mirror-image method

This method was the first computer method for room-acoustics modelling which has been used together with a fully functioning auralisation component.¹⁴ The method is inherently deterministic and allows for exact

calculation of room impulse responses the length of which is limited by available computer power.

The fundamental idea of the mirror-image method is as follows. With a sound source being located in front of an unbounded rigid wall, the sound field at any point in front of the wall can be calculated by summing up the sound field caused by the original source at this point and the sound field caused by a secondary source. The secondary source is specified by mirroring the original source at the plane of the reflecting wall. With more than one surface being present, multiple reflections will occur and, consequently, there will be multiple-order mirror-image sources. The implementation is easy in the case of rectangular rooms,¹⁵ but requires more complex consideration in the case of arbitrary room shapes,¹⁶ amongst other things because visibility tests for obstruction must be performed. The ratio of computed image sources to valid image sources, i.e. the percentage of images that pass the tests, increases exponentially with the order of reflection, thus making the algorithms very inefficient for room simulations including high-order reflections.

The advantage of the mirror-image method is that all mirror images constitute virtual sound sources, which are well-adapted to the sound-field representation desired.

An interesting aspect of the application of the mirror-image method to rectangular rooms, where it is most frequently used, is that not all image sources are visible. Already at the order of two the number of hidden-image sources is equal to the number of visible-image sources, and it is much higher for higher-order image sources. However, it can be shown that from a subset of image sources being located in the same position, exactly one is visible, which is determined by the position of the receiver in the room. In such a way the visibility test can be skipped in these cases. This kind of consideration is the subject of current scientific discussion in the context of the role of hidden image sources.⁴⁷

3.2 The ray-tracing method

Since Krokstadt *et al.*¹⁷ introduced ray tracing for the prediction of the acoustical response of rooms, a variety of different implementations of this method have evolved. In ray tracing, sound sources emit certain 'tokens' into all spatial directions, the tokens being rays, particles (e.g. ref. 18), conical beams¹⁹ or pyramid-shaped beams.²⁰ The tokens obey the principle of Fermat. As air can be considered as a homogeneous, isotropic medium at rest, the tokens propagate along straight lines and are geometrically reflected when hitting a reflecting surface. Tokens which arrive at the position of the sound receiver, given that predescribed detection conditions

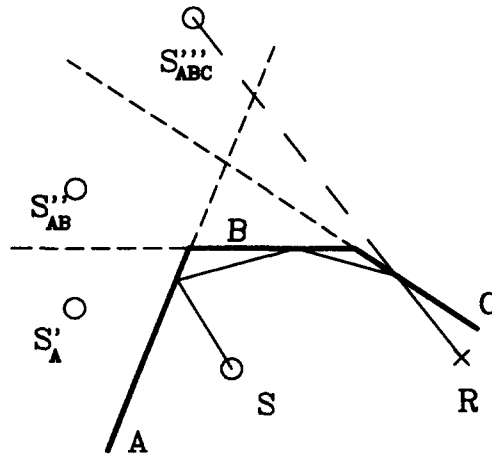


Fig. 5. Construction of the same virtual source by the mirror-image-source method or by ray tracing. The virtual source S_{ABC} can be found by mirroring the source S at the walls A , B and C or by tracing a ray in the corresponding direction. The virtual source can be found by moving from the receiver R against the direction of incidence of the ray. The distance between the virtual source and the receiver equals the total ray length.

are fulfilled, contribute to the room-response quantity to be determined for the position of the receiver.

In general terms, the tokens are carriers of physical properties (e.g. energy or wave form) or they just serve to determine meaningful paths of wave propagation from which virtual sound sources can be reconstructed.^{21,22} See Fig. 5 for an example of such a reconstruction.

The different interpretation of the tokens has led to two different models both based on the ray-tracing algorithm. The conventional method may be called *energy tracing* whereas the sound path finding may be referred to as *hybrid ray-tracing*, since it uses ray-tracing algorithms in combination with the concept of mirror-image sources. Given that systematic errors, owing to the finite volume with respect to the area of the target, are corrected,²³ the virtual sources as determined by ray tracing are identical to those being rendered by the mirror-image method. As a matter of fact, both methods would arrive at exactly the same results in the case of unlimited computational effort.

3.2.1 Conditions for detection

It is a fundamental problem of ray tracing that neither an ideal ray nor a point-like particle can hit a point-like receiver due to their infinitely small size. To fix this problem, finite dimensions must be assigned to the target and/or to the tokens to be traced.

Different solutions have been proposed and implemented to deal with this

problem, the most common one being that a finite target volume, in the form of, for example, a sphere, a cube or a disk, is used as a ray detector.^{18,24-26} It should be recalled in this context that only spheres are isotropic. Other receiver forms may result in anisotropic response patterns such as ripples in the directional characteristics in the case of a cube, or dipole characteristics in the case of a disk.

In the case where finite dimensions are assigned to the tokens traced, as, for example, in cone or pyramid tracing, there may be an infinitely small (point) target, but the tokens traced are of finite extensions such as to cover a given section of solid angles. Then, for example in the case of cone or pyramid tracing, the centre ray is taken to be detected whenever the bottom-plane of a cone or pyramid hits the target point.^{19,20,27} In the case of cone tracing, overlapping cones are handled by appropriate energy correction; in the case of pyramid tracing this is not necessary when the pyramids are arranged such as to completely cover the surface of a sphere with non-overlapping triangles of approximately the same size.

When targets of finite and constant size are applied, the size must be adapted to the ray density used, to ensure detection. Otherwise, rays travelling along valid sounds paths may simply miss the receiver when the distance of adjacent rays exceeds the size of the receiver. Since the rays are relatively dense in space close to the sound source, targets of constant size tend to lead to multiple detections of the same valid sound path. Figure 6 gives an example of this effect.

The multiple-detection effect is an outcome of the propagation characteristics of spherical waves and within the energy-tracing system it is used to model these characteristics. It is, however, preferable to reduce the

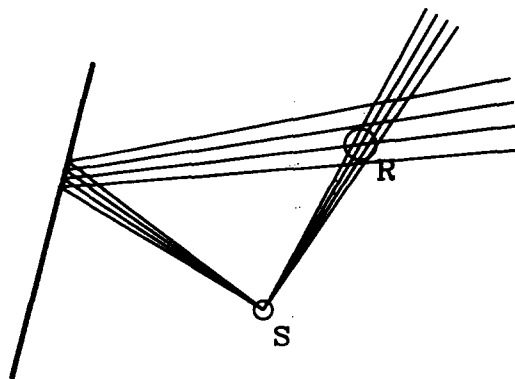


Fig. 6. Two-dimensional example of a multiple detection of the direct-sound path in the ray-tracing method with a target of constant size. The detector is hit by four rays representing the direct sound. The reflection of the wall is detected only once. The angular difference of the rays is constant.

energy of the rays in reciprocal proportion to the length of the rays rather than to reduce their number with distance.²⁸ As a matter of fact, it is only this method, which can be applied within hybrid ray-tracing detections, which would lead to a spatial blurr of the image sources, depending upon the parameters of the algorithm.

In any case, the use of rays as well as detectors will always induce errors specific to the geometrical model of sound propagations.^{21,28} These errors are systematic ones which can be corrected for by reconstructing virtual sources by mirroring the original sound source with respect to all walls which are hit by rays. In this way, the 'authentic' positions of image sources can be determined. The validity of image sources determined in this way can be checked by applying visibility and obstruction tests as known from the mirror-image-source method or by a back-tracing process,²³ allowing for targets of arbitrary size, provided that a minimum size is given which results from the resolution aimed at in terms of solid angle and ray length. Even detectors of infinite size can be used with this method.⁴⁸

For spherical detectors, the minimum radius $r_{\min}$ is given by

$$r_{\min} = l(2\pi/N)^{1/2} \quad (1)$$

where N is the number of rays and l the ray length. Since $r_{\min}$ is a function of the ray length, it appears to be an advantage to use detectors of variable size accordingly. In the case of spherical detectors with the radius $r_{\min}$, the results would correspond roughly to that of the cone-tracing method.

3.2.2 Spatial resolution

One of the most critical parameters of ray tracing is the appropriate number of rays to be applied, a number which is often estimated on heuristic rather than on systematic grounds. To increase the quality of this estimation we shall, in the following, consider the adequate number of rays on the basis of room geometry.

The probability for a particular reflection to be found with a ray-tracing algorithm is a function of both area A_i of the reflecting wall and the angle of incidence α_i . By defining the area which is 'seen' by the ray upon incidence as the effective area S_i , one gets

$$S_i = A_i \cdot \cos(\alpha_i) \quad (2)$$

As the number of reflections may be very high, an approach offers itself in which A_i , α_i and S_i are considered as stochastic variables. Under the simplifying assumption that all reflections are statistically independent, the probability-density function of A is solely determined by the areas of the reflecting surfaces in a room but not by their geometrical arrangement. A can

only take on discrete values, its probability-density being composed from Dirac impulses as follows.

$$f_A(A) = \sum_i \delta(A - A_i) \cdot A_i / A_T \quad (3)$$

Here A_i are the areas of the individual walls and A_T their sum. In a diffuse sound field the probability density of the angles of incidence is given by

$$f_\alpha(\alpha) = \begin{cases} \sin(2\alpha) & 0 < \alpha < \pi/2 \\ 0 & \text{elsewhere} \end{cases} \quad (4)$$

Considering eqn (2), one arrives with eqns (3) and (4) at the probability density $f_S(S)$ of the effective area, S , such that

$$f_S(S) = \sum_i \begin{cases} 2S / (A_i \cdot A_T) & 0 \leq S \leq A_i \\ 0 & \text{elsewhere} \end{cases} \quad (5)$$

The mean value of this term can be derived as the first moment of $f_S(S)$ and is given by

$$S_{AV} = \int_{-\infty}^{\infty} S \cdot f_S(S) \cdot dS = 2 / (3 \cdot A_T) \cdot \sum_i A_i^2 \quad (6)$$

S_{AV} is, at the same time, the mean effective area for any reflection—determined solely by the room geometry. In Fig. 7 we show measured as well as calculated values for a particular auditorium at our university. The ‘measured’ values stem from a ray-tracing process in the auditorium using a CAD file of architectural data from which a histogram of the effective areas was derived. The resulting relative frequencies were divided by the respective class intervals. The graphs show good agreement of measured and calculated values. This has been verified for a number of rooms of different form and geometrical complexity.

The forms of $f_S(S)$ cannot easily be interpreted; however, they are typical for their respective rooms and they do not at all correspond to a Gaussian distribution. To estimate the detection probability p , we first have to introduce a detection procedure as a function of the ray length l . Consequently, with the use of eqn (5), the probability-density function of p can be calculated—the first moment of which, p_{AV} , comes out as being the mean detection probability of a reflection to be found with a ray of length l . One thus arrives at

$$p_{AV} = \sum_i A_i / A_T \cdot \begin{cases} \frac{2}{3} \cdot A_i / \Delta\Omega l^2 & A_i \leq \Delta\Omega l^2 \\ (1 - \frac{1}{3} \cdot (\Delta\Omega l^2)^2 / A_i^2) & A_i \geq \Delta\Omega l^2 \end{cases} \quad (7)$$

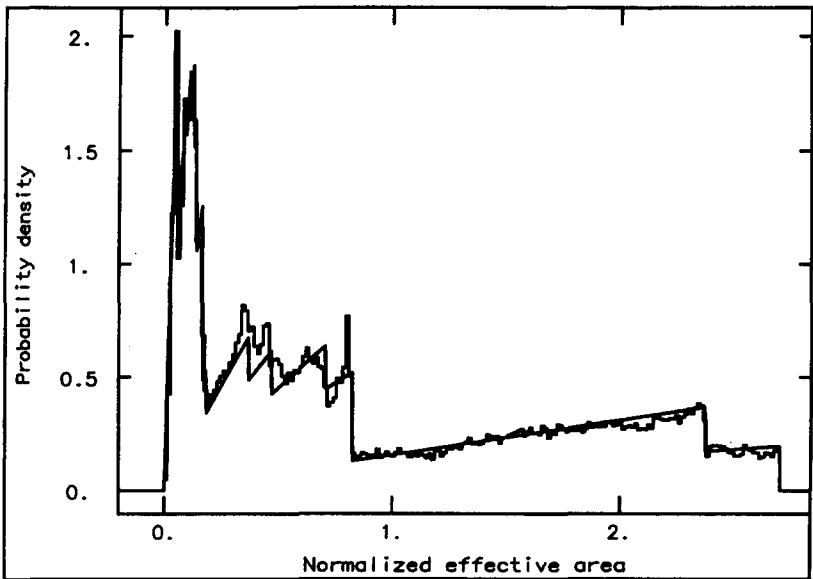


Fig. 7. Probability-density function $f_S(S)$ of the effective reflecting area of a lecture hall. The effective area, S , normalized so that unity corresponds to $S = S_{AV}$, which is the mean value as derived from eqn (6). The step curve represents measured values and the straight lines theoretical ones as derived by eqn (5).

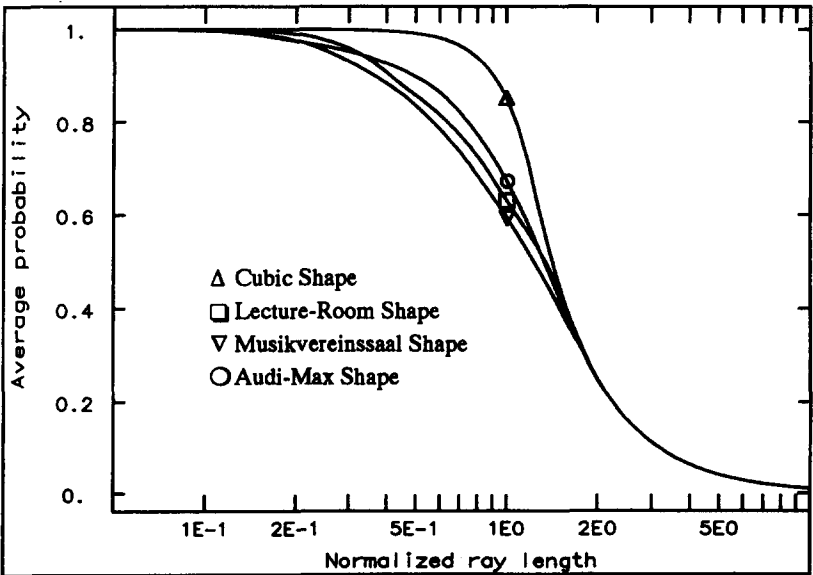


Fig. 8. Mean detection probability as a function of the standardised ray length for four different rooms. Standardisation was performed in such a way that unity corresponds to that ray length where the area covered by one ray equals the average reflecting surface of the room, i.e. $\Delta\Omega/l^2 = S_{AV}$.

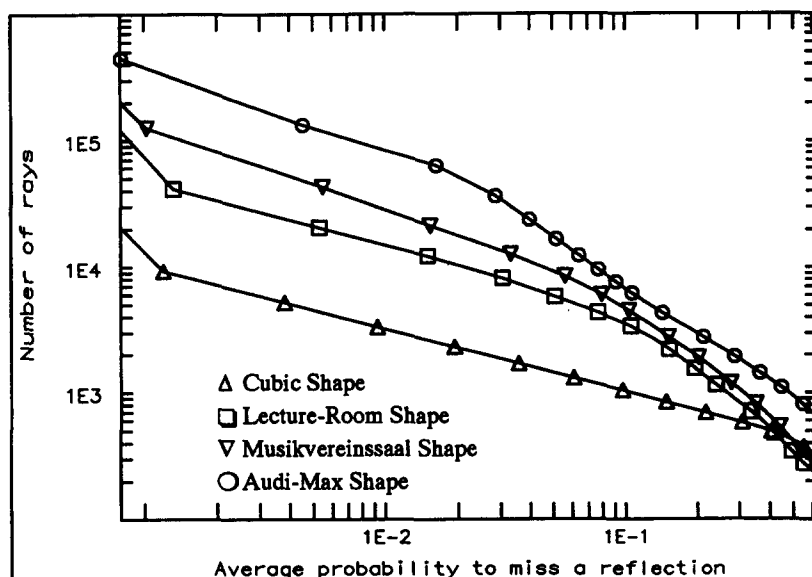


Fig. 9. Necessary ray number as a function of the probability to miss the target, $(1 - p_{AV})$, plotted for a ray length of ten times the mean-free-path length for the four different rooms as used above. To allow for double-logarithmic plotting, the abscissa is used to denote the mean probability for missing the target.

with $\Delta\Omega$ being the solid-angular interval which is covered by a ray, as can be calculated from the ray number through $\Delta\Omega = 4\pi/N$. In Fig. 8, p_{AV} is depicted for a number of different rooms.

With a given mean detection probability, it is possible to calculate the necessary ray number by using eqn (7) in inverse form. Figure 9 gives the results for the four rooms used above.

It becomes clear from the figure that one would need 3000 rays for a cube-shaped room to achieve a detection probability of 99%, whilst in the most complex room amongst our samples, the AudiMax, about 200 000 rays would be required.

Looking at the problem the way we did, the results concern single reflections. Yet, there are most often multiple reflections involved in the creation of a virtual source. Assuming, as a first-order approximation, that there is a proportionality of the detection probability and the ray length between the first and the 10th reflection, one can state the following for the example above. For a detection probability for the 10th-order reflection of 99%, the probability to find the corresponding virtual source is 95%. For a detection probability of 90%, the chance to find the virtual source drops to 50%. This relation gets even worse for virtual sources of higher order.

If one further considers the fact that natural rooms tend to show

reverberation times corresponding to a length of the order of 40–80 times their mean-free-path length, one realises that immensely high numbers of rays would be necessary for a straightforward determination of the late parts of impulse responses of such rooms on the basis of ray tracing. It is for this reason, that special algorithms are required for this task, such as are discussed in the next section.

4 THE MODELLING OF REVERBERATION

The generation of the late part of room responses, i.e. the reverberation part, is still not feasible in terms of computational effort required, nor would it make sense on perceptual and/or physical grounds. Traditionally, reverberation is treated by stochastic methods in room acoustics. Therefore it appears to be reasonable to describe reverberation in terms of stochastic parameters in room-acoustical modelling as well.

However, there is one fact that must be considered: although the reflection density in the reverberant component of a sound field is high enough not to allow the auditory system to selectively discriminate the individual arrivals, the reverberation part still gives rise to very typical perceptions such as reverberance, room impression, spaciousness and, eventually coloration. These are perceptions which are typical for a particular room, and, consequently, must be modelled accordingly. Further, the transition between the 'exactly' and 'stochastically' simulated parts of room responses constitutes a problem.

At the present time there are a number of different approaches for the generation of reverberation in the context of sound-field modelling by computer. One of them starts with the concept of a stochastic procedure which generates individual reflections of medium order. High-order reflections, i.e. the late reverberation tail, are then added by a direct waveform-generation process which is controlled by the same stochastic parameters.^{29–31} In another procedure, reverberation is modelled as a Poisson process.³² Further possibilities are established by using a low-resolution ray-tracing procedure and controlling the energy of detected rays according to their inverse detection probability.^{28,33}

Since in the late reverberation component of the sound field of practically all rooms diffusely reflected sound energy exceeds geometrically reflected sound energy, it is reasonable to consider models for the generation of reverberation which are based on diffuse reflection or, at least, a mixture of geometrical and diffuse reflections. There is one algorithm known from computer graphics, namely 'radiosity', which has been investigated in this context.²⁰ However, there is as yet no successful implementation of it available as far as binaural room simulation is concerned.

5 CRITERIA FOR TRUNCATION

To arrive at meaningful computer simulations of a sound field with reasonable effort, the simulation algorithms must be terminated in time somewhere. This becomes immediately clear when one thinks of algorithmic problems such as a ray being caught between two parallel walls. Also the transition from deterministic to stochastic algorithms requires the timing of the truncation of the former and the start of the latter.

For accuracy with respect to the authenticity of a simulation, the criteria used to determine the point of truncation are certainly of some relevance. In principal, truncation criteria can be based on the length of the room impulse response, on the energy, on the maximum order of reflection to be considered or on a combination of these.

Although the maximum order of reflection is obviously an important criterion, it does not suffice in general. The arrival time delays of reflections of the same order with respect to the direct sound are of approximately Gaussian distribution.³⁰ The same holds for the distribution of ray energies. The energy of individual rays decreases with the time elapsed, but at the same time the temporal density of reflections increases. An absolute criterion in terms of order of reflection or of energy is, thus, difficult to establish. Amongst other effects, a particular reflection which stands out in time, comes from a 'sensitive' direction or shows particularly audible spectral characteristics, may be individually perceptible although its total energy is comparatively low.

The maximum length of the deterministic part of the impulse of a room seems to be a more appropriate criterion for truncation of the deterministic simulation procedure. Amongst other reasons, such a criterion can be based on perceptual evidence and also be estimated from the reverberation time of a room. If energy is not considered in this context, one has the advantage that the calculation of virtual sound sources need not be repeated every time the acoustical treatment of a room, i.e. the absorption coefficients of the walls and other absorbing items, is modified. Only the spectra and not the position and orientation of the virtual sources need new consideration in these cases.³⁴

6 TREATMENT OF WALL REFLECTIONS

Within the geometrical model, a wall can be described as one-dimensional linear system. The transfer function of that system can be defined as the ratio of the spectrum of the reflected sound to the spectrum of the incoming sound and is given by the reflectance of the surface material. The reflectance is of

complex value and dependent upon the angle of incidence and on frequency. The wall can be modelled as a parametric linear filter, with the angle of incidence as the parameter.

Unfortunately, it is not very easy to measure the required reflectance data for a wall, namely if an already existing environment has to be simulated and in-situ measurements are required. Only few measurements have been taken for frontal³⁵ and arbitrary incidence. Widely available in the literature is the spatially averaged absorption coefficient as measured in a reverberation chamber. However, if highly absorptive materials are measured, sound travelling parallel to the absorptive surface is much less attenuated as sound travelling orthogonal to that surface and spatial diffusivity is lost (see e.g. Ref. 36). That leads to principal errors, e.g. measurements of absorptions coefficients greater than unity occur frequently and the reproducibility of the method is generally bad. Nevertheless, due to its easy access, the averaged absorption coefficient measured in octave or third-octave frequency bands is still widely used in binaural room-simulation systems.^{34,37,38}

The frequency dependence has induced two different types of simulating system, broad-band and narrow-band models. In the narrow-band systems, the whole simulation process is split up into different frequency bands, usually octaves.^{14,24} The rays or image sources carry an initial energy which is reduced at every reflection according to the absorption coefficient of the surfaces. Alternatively, within energy-tracing systems, a random number can be chosen to decide if the ray or particle is completely absorbed or left unchanged. The probability of the ray to be absorbed is proportional to the absorption coefficient (for a comparison of both methods see Ref. 13). The broad-band solution is obtained by filtering the result of every frequency band with an ideal band-pass filter and by subsequent superposition.^{14,49}

In broad-band systems, the wall may be described by its impulse response. The ray or the image source carries an initial signal which is convolved with the impulse response of the wall for every reflection.^{1,38} The impulse response of the wall can be obtained from the absorption coefficients as follows. The magnitude of the reflectance can be computed from the absorption coefficient for some discrete frequency. The magnitude for all frequencies can be obtained by linear or spline interpolation. Applying Hilbert transforms to the logarithm of the magnitude yields a minimum phase distribution for the given magnitude function. The impulse response can be calculated by inverse Fourier transforms. Alternatively, linear phase filters or non-minimum phase filters can be created by choosing appropriate filter functions. Figures 10 to 12 show the transfer function and the impulse response of the resulting filter for a spline interpolation of the magnitude, for a spline interpolation of the level and for the resulting filter in a narrow-band system for one selected surface material.

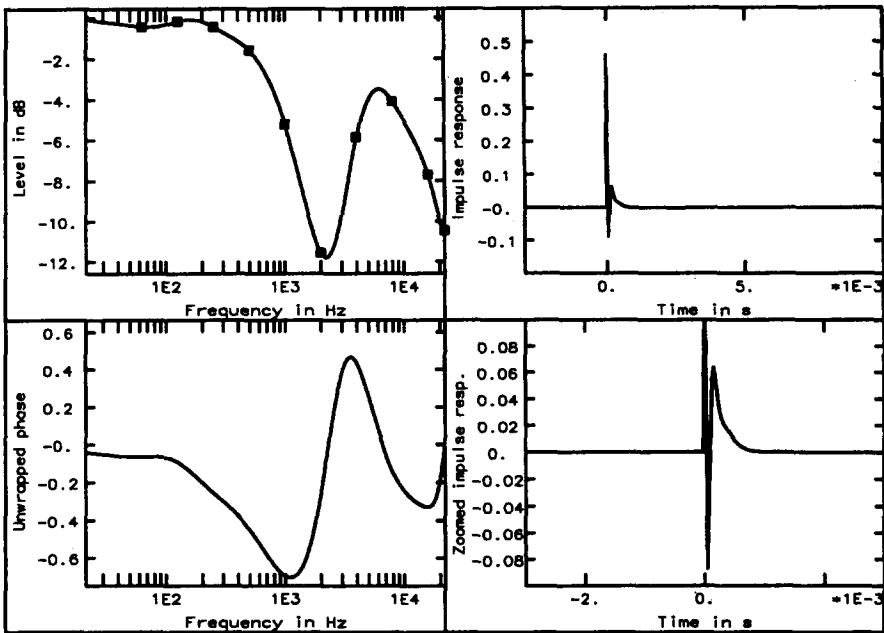


Fig. 10. Transfer function and impulse response of a specific wall material. The transfer function has been obtained by interpolation of the reflectance magnitude.

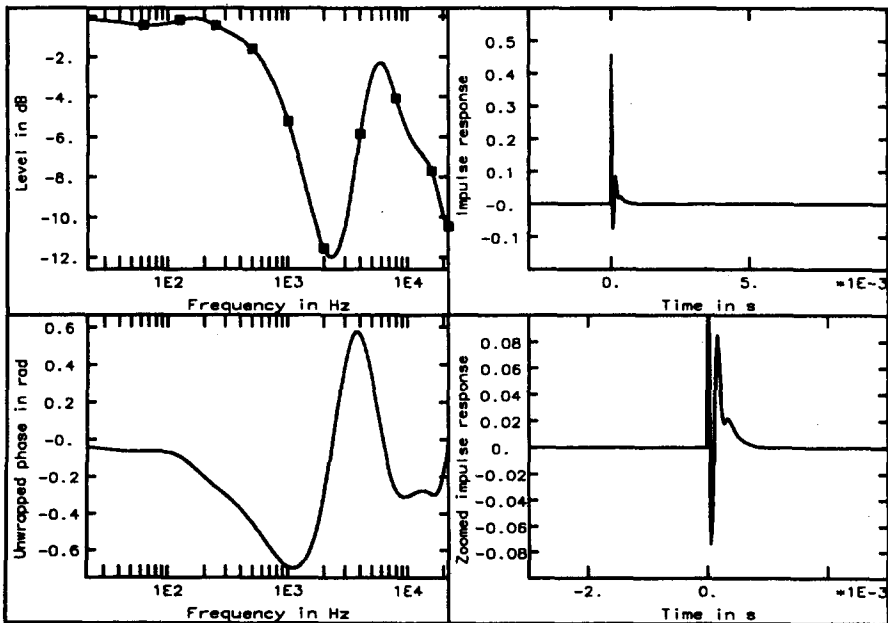


Fig. 11. Transfer function and impulse response of a specific wall material. The transfer function has been obtained by interpolation of the reflectance level.

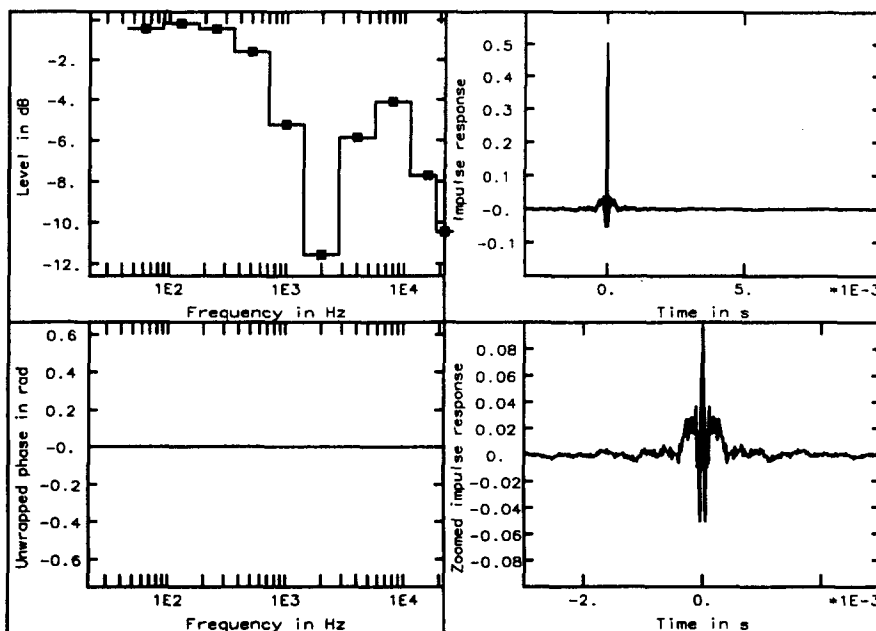


Fig. 12. Effective transfer function and impulse response of a specific wall material obtained by simulation in octave bands.

Generally speaking, the interpolation of the level causes more ripple in the transfer function than does the magnitude interpolation. The magnitude of the resulting transfer function must be limited to the interval $[0, 1]$, to avoid active walls. Owing to the spline interpolation, clipping effects may occur. This will occur more frequently, if level interpolation is used. Simulations based on octave bands cause the magnitude of the effective reflectance to be a step function. The steps lead to ripples in the impulse response. owing to the zero phase of the filter, the impulse response is non-causal. It is not yet clear whether any of the effects mentioned cause auditorily relevant effects. Recent investigations³⁹ suggest that phase treatment is not very critical.

7 MODELLING OF DIFFUSE REFLECTIONS

As stated above, one of the major drawbacks with the principles of geometrical acoustics is the neglect of wave-propagation effects, namely of diffuse reflections. On hitting a wall, sound energy is usually reflected in many if not all directions. To model this fact, however, extension of the pure geometrical model is necessary.

The diffusion coefficient of a wall can be defined as the ratio of the energy

which is not reflected according to the reflection law to the total reflected energy. Wall reflection can approximately be considered in terms of two components, one rendering ideal specular reflection and the other one ideal diffuse reflection—i.e. part of the reflected energy is bounced back according to the geometric law of reflection and the rest is returned according to Lambert's law, which holds for ideal diffuse surfaces. The respective parts of the energy can be determined from the diffusion coefficient and the absorption coefficient.

There are several ways to model reflection at an ideally diffuse surface. In the energy-tracing method, the direction of the reflected ray can be chosen by a random process, where the probability density of the respective direction is determined by Lambert's law. In this case it would even be possible to adapt the random process to the spatial diffusion properties of the surface—if known. Figure 13 shows a calculated scattering pattern of a rectangular surface.

The random model for geometric reflections, as mentioned above, can obviously be modified easily to include diffusion in this way. Yet the number of rays has to be increased significantly to obtain valid results on a statistical basis.

Another possibility is to split the ray at the point of reflection and trace each of the new rays separately. In the mirror-image-source method, a reflection at a diffusive surface would thus produce a cloud of virtual-image sources from each individual image source.

In general, all these methods lead to a drastic increase in the computation time required. For this reason, these authors favour a split-up of the simulation process into two parts as follows.

A first process treats all surfaces as being ideally specular. This process may respond to some optimisation ideas as described above. The second

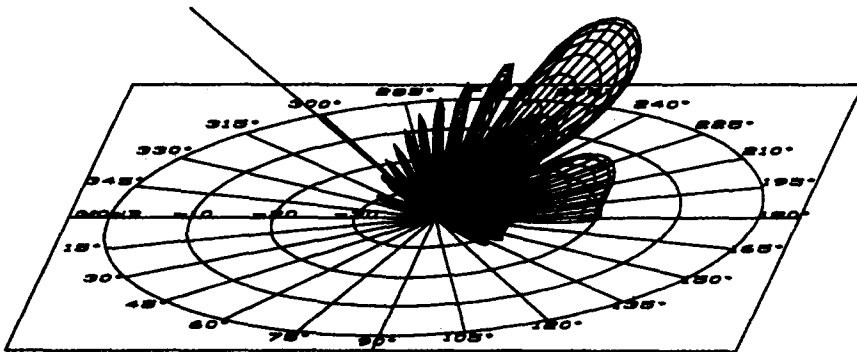


Fig. 13. Calculated scattering pattern of a square-shaped surface including edge diffraction. Direction of incidence is 45° parallel to an edge of the square. The wavelength is about one-sixth of the edge length.

process considers all walls as reflecting diffusely in an ideal sense. Simulation can then be based on stochastic methods, e.g. radiosity.²⁰ In both parts the energy reflected to the other kind of reflection would be treated as being absorbed. Upon sound being reflected at diffusive surfaces, sound energy is distributed over a far wider solid angle than by a specular reflection. Thus the intensity of the reflected sound in any direction is much lower than that in the main direction in the specular case. Diffuse reflections are, consequently, very hard to localise. It may be assumed that modelling of diffuse reflection requires much less accuracy than the modelling of specular reflections.

To be sure, the results of both simulation processes, the geometrical and the diffuse one, must finally be added again either by combining the obtained distributions of virtual sound sources, or simply by summing up the impulse responses at the listeners' eardrums.

8 TREATMENT OF SOUND-SOURCE DIRECTIVITY

A sound source can generally be described by its directional characteristics. One possible representation of the far-field directional characteristic of a source is the complex sound-pressure distribution on a sphere centered around the acoustical centre of the source. Restriction to the far field seems reasonable, since in room acoustics the distance between virtual sources and receivers is usually large. As far-field characteristics do not depend upon distance, one can consequently restrict the representation to two dimensions (azimuth and elevation). Thus the amount of data required to model the source characteristics is significantly reduced.

Literature may render relevant data for various natural sound sources (e.g. Ref. 40) and manufacturers of technical sound sources tend to supply their products with detailed directivity data. An example for measured data of very high resolution can be found, for example, in Ref. 50. However, no phase data are usually available and, especially in the case of natural sound sources, these are difficult to measure.

For these reasons, similar to the treatment of wall reflections, zero-phase or minimum-phase functions must be generated. From full sets of spectral data, impulse responses can then be obtained by inverse Fourier transforms for every direction of emittance.

During the simulation process, the orientation of virtual sources with respect to the original source can be maintained by assigning initially a local coordinate system to the sound source which is mirrored or reflected at each reflection. The direction of emittance for a virtual source is determined with respect to the position of the receiver through appropriate coordinate transforms. Signals being emitted by a source will be filtered with the specific impulse response for the corresponding direction of emittance.

In the ray-tracing method, a further possibility for modelling the

directivity of sound sources is given, since the ray density in each spatial sector can be chosen in accordance with the magnitude of the directivity, as averaged over that interval. The ray density will thus be high in such sectors into which the source emits high energy, etc. The drawback of the method is that phase information is lost. Since the directivity generally depends largely upon the frequency, this method is restricted to narrow-band simulation systems or very simple sources.

The authors therefore suggest the following description of sound-source directivity. Directivity can be modelled by means of so-called directivity filters. These filters describe the sound spectrum emitted in a specific direction with respect to a reference direction. The directivity filter for the reference direction equals unity by definition. One possible interpretation of this concept is that the spectrum emitted in the reference direction can be considered as being the transfer function of the sound source, and the directivity filters describe the variation of the emitted sound with the direction of emittance. For natural sound sources such as human speakers or musical instruments it is not possible to define a transfer function. However, the anechoic signals have to be recorded in some direction of emittance, and thus the recorded signals already carry an initial transfer function. If the recording direction is chosen to be the reference direction, the directivity filters contain all information needed to model the directional characteristics of the sound source. Recalibration to a different recording direction is also possible.

Measurement of directivity filters can be performed by simultaneously measuring the emitted sound spectra in the direction under test and the reference direction, and dividing the results. Also, the phase characteristics of the directivity can be captured this way. This measurement technique can even be applied to natural sound sources, since the measurements are independent of the emitted signal, provided that the signal-to-noise ratio in the reference direction is sufficient for all frequencies of interest. In cases where the source is only able to emit narrow-band signals, directivity filters can be obtained by measuring, for example, the third-octave spectra and dividing these by the third-octave spectrum measured simultaneously in the reference direction. To be sure, phase information is lost this way but a minimum-phase function can be added later to create directivity impulse responses. Figure 14 shows, as an example, measured directivity filters of a loudspeaker.

Currently, the authors favour an approach where the directivity data are described on the basis of a mathematical model which can be based on synthesis by superposition of orthogonal elements like multipoles or spherical functions. As an example, Fig. 15 shows the magnitude of the measured directivity for 4 kHz. Applying a two-dimensional Fourier transform with respect to the azimuth and elevation angles to that data gets

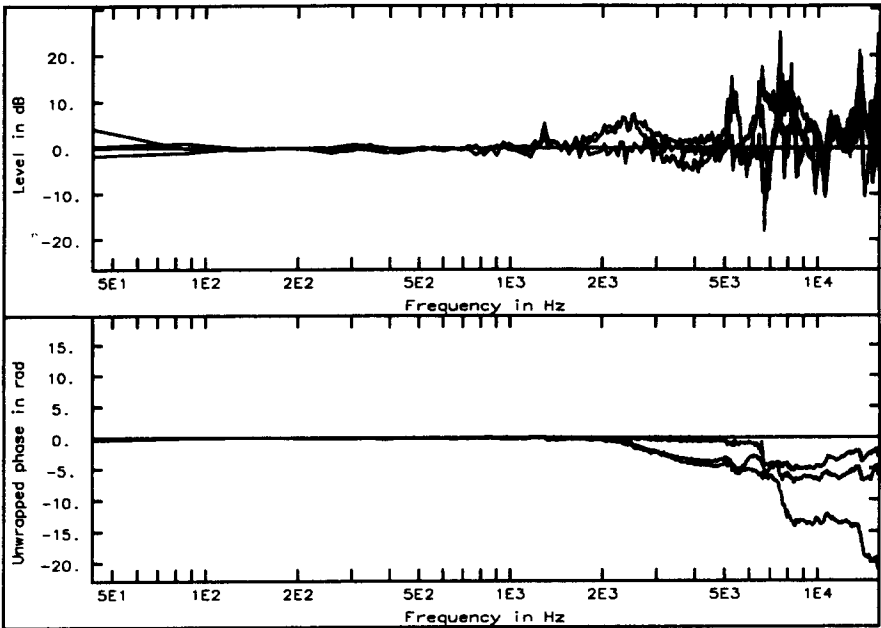


Fig. 14. Transfer functions of measured directivity filters for the azimuthal directions of 0, 90, 180 and 270°. The filter for 0° is flat by definition.

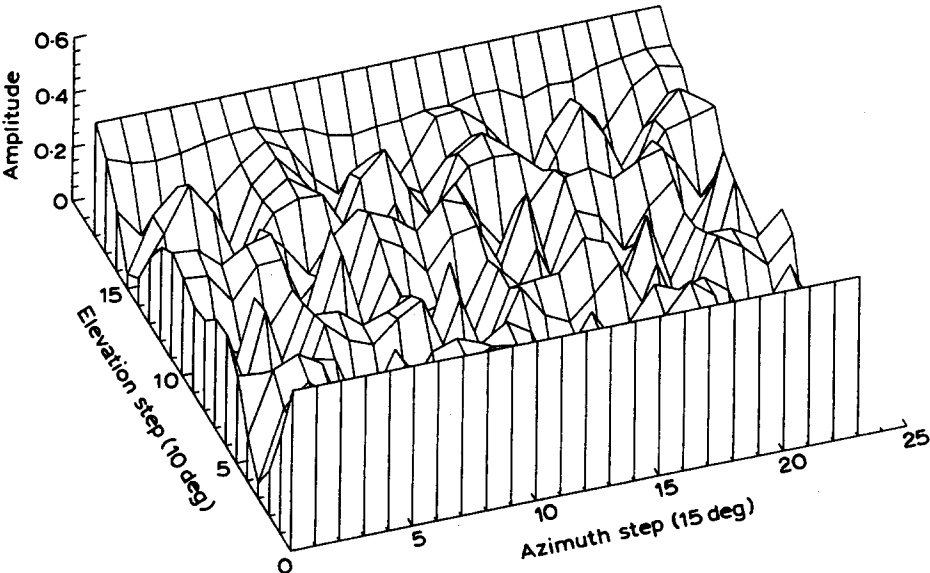


Fig. 15. Measured magnitude of the directivity of a loudspeaker as a function of azimuth and elevation for a frequency of 4 kHz.

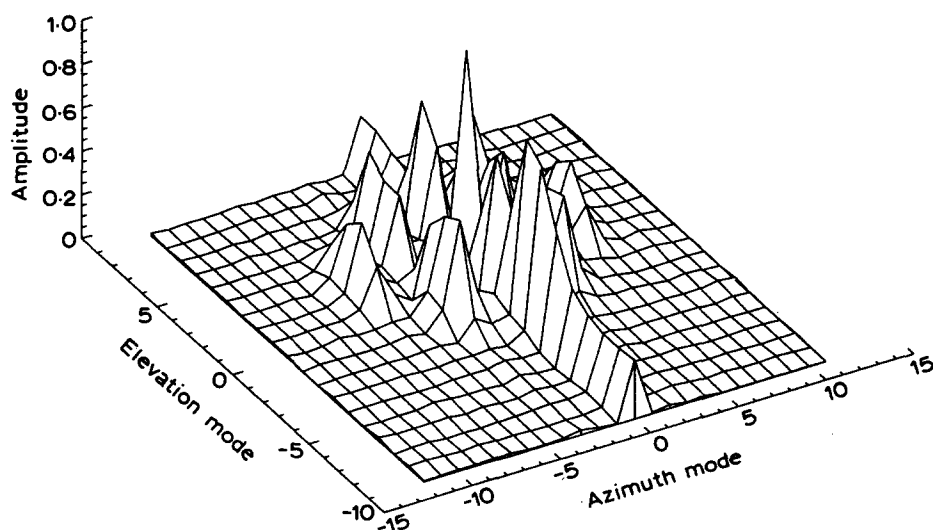


Fig. 16. Magnitude of the two-dimensional Fourier-transforms with respect to azimuth and elevation of the data depicted in Fig. 15 as a function of the angular frequency.

a description in the angular frequency domain, as can be seen in Fig. 16. The description in the angular frequency domain shows a less-complex pattern and the amount of data can be reduced.

This kind of description has to be created for each frequency, and consequently each angular mode is a function of frequency. However, initial experiments suggest that interpolation across frequency in the angular-frequency domain is much easier than in the angular domain and, for most angular modes, only a few supporting points need to be stored.

9 AURALISATION

One of the main advantages of the concept of virtual sound sources is its usability for auralisation, i.e. the results of the simulations can easily be made audible to human listeners, provided that the position and the specific linear filter of each virtual source are known. It may be recalled that the linear filters include the information on all wall reflections involved in the formation of the virtual source, and on the source directivity, for a specific direction of emittance.

9.1 Computation of binaural impulse responses

For auralisation the impulse responses at the ear drums of the two ears of the listener must be determined (the so-called binaural impulse responses). To

obtain these, the primary and all virtual sound sources are simultaneously excited by a Dirac impulse, filtered by the respective individual linear filters, and propagated to the receiver. During propagation the signal is attenuated due to the $1/r$ law and to air absorption—and delayed. Air absorption is frequency-dependent. Roughly speaking, air is a low-pass filter whose magnitude of the transfer function has a Gaussian shape and whose cutoff frequency depends upon the distance travelled and on humidity.⁴¹ At high frequencies, air is even dispersive, yet this effect can be neglected here as the

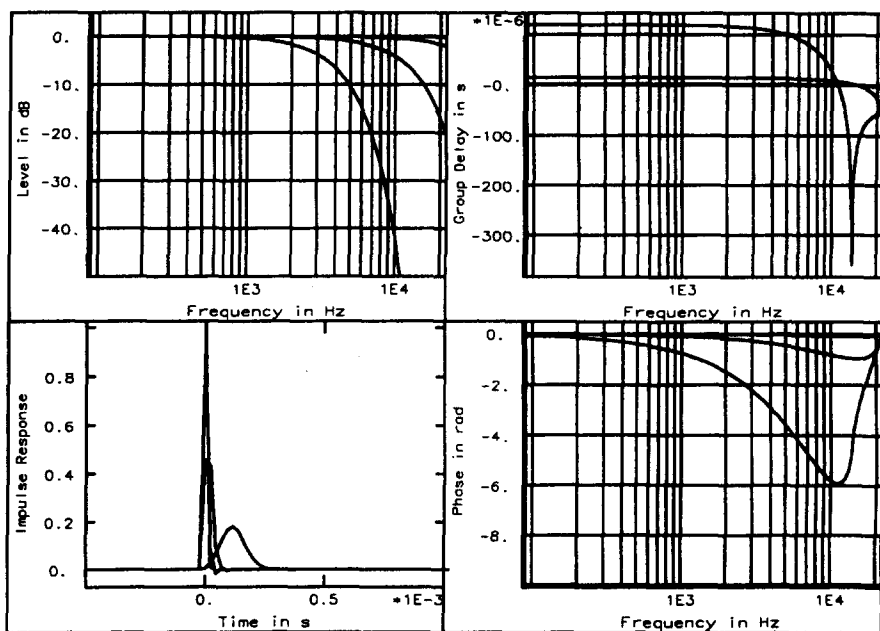


Fig. 17. Air-absorption filters for distances corresponding to 1, 10, 100 and 1000 ms. The figure shows level, group delay and phase as a function of frequency as well as the resulting impulse responses.

relative differences of maximum and minimum values of the speed of sound is only about 4.5×10^{-4} in the audible frequency range.⁴² Air can thus also be modelled as a zero-phase or minimum-phase filter. Figure 17 shows the resulting filters for different distances.

The binaural impulse responses of the environment are determined by filtering the signals stemming from all primary and virtual sound sources arriving at the listener's position with the listener's so-called *head transfer function* (HTF) for each corresponding direction of incidence. Summing up the contributions of all sound sources finally yields the binaural impulse responses as desired.

9.2 Measurement and equalisation of head transfer functions

Binaural room simulation is based on the idea that a listener, when being supplied with sound signals at his/her eardrums which are coined after those occurring in a real environment, will have the same auditory events as he/she would have in the respective real environment. To realise such simulations, the HTFs of the listener must be known.

HTFs can be obtained by appropriate direct acoustical measurement. In measurements most often performed in anechoic chambers, the following practical simplification is common (e.g. Ref. 43). One can state that the head of a listener is in the far field of the direct and virtual sound sources in all practical cases. Thus, it can be assumed that the incoming waveforms are approximately plain waves and, consequently, the HTFs can be taken as being independent of distance, except for a frequency-independent magnitude factor.

The reproduction of binaural signals is preferably performed via headphones. Sounds which are emitted from these, prior to arriving at the ear drums, travel along the ear canal, resulting in linear distortions. Further linear distortions when measuring HTFs result from the fact that the measuring microphone is not positioned directly in front of the ear drum, but in a reference plane somewhere in the ear canal.

However, space propagation of sound in the ear canal is unidimensional, both kinds of linear distortions can easily be compensated for by linear equalisation. For simplicity we consider only one of the two ears in the following. The results, nevertheless, hold for both ears.

Given that the microphone for the measurement is positioned at a distance d from the eardrum, the measured transfer function $M_1(\omega)$ is given by

$$M_1(\omega) = H_{\phi\theta}(\omega) \cdot E_{d,z,1}(\omega) \quad (8)$$

where $H_{\phi\theta}(\omega)$ is the desired HTF for the direction as given by the azimuth ϕ and the elevation θ , $E_{d,z,1}(\omega)$ being an error function dependent upon the displacement d and the mechanical impedance Z , terminating the ear canal towards the middle ear. The latter will certainly be affected by the microphone.

When measuring the transfer function of a headphone $H_h(\omega)$ with the same set-up, one gets

$$M_2(\omega) = H_h(\omega) \cdot E_{d,z,2}(\omega) \quad (9)$$

where $E_{d,z,2}(\omega)$ is a second error function.

In cases where the radiation impedance as seen from inside the ear canal towards the external sound field is the same for both the headphone and

loudspeaker measurement—as can be assumed in the case of acoustically open headphones—the two error functions are identical and one arrives at

$$E_{d,z,1}(\omega) = E_{d,z,2}(\omega) \quad (10)$$

When reproducing the signal of one specific virtual sound source, the spectrum of the sound pressure signal at the ear drum, $P(\omega)$, is given by

$$P(\omega) = M_1(\omega) \cdot H_h(\omega) \cdot S(\omega) \quad (11)$$

where $S(\omega)$ is the spectrum of the signal that the virtual sound source would cause at the listener's position with the listener absent. $P(\omega)$ thus includes both the error function and the headphone transfer function. By multiplying $M_1(\omega)$ with the inverse of $M_2(\omega)$ one obtains

$$P(\omega) = H_{\phi\theta}(\omega)S(\omega) \quad (12)$$

which is exactly the desired result for the simulation intended. Consequently, the inverse of $M_2(\omega)$ is the correct equalisation function. The particular equalisation required depends upon the type of headphones used. For example, the use of so-called diffuse-field-corrected headphones prompts diffuse-field equalisation, and the use of free-field-corrected headphones prompts free-field equalisation of the binaural signals prior to auralisation. Equalisation can be performed in two steps after the binaural impulse response has been computed, the second step rendering the adaptation of the type of headphone. Such an equalisation procedure has the advantage of only relative measurements being required, i.e. neither absolutely calibrated microphones nor a reproducible placement of the microphones are necessary.

10 SOME CONSIDERATIONS CONCERNING REAL-TIME PERFORMANCE

One of the most challenging aspects of virtual auditory environments is interactivity, i.e. the subject should be able to move and act inside the virtual environment. To achieve this ultimate goal, the system must be able to react on events produced by the subject or by modifications in the virtual environment itself—with delays that are below the respective perceptive thresholds. This could be called a true real-time performance; however, this condition is hard to fulfil. Real-time realisation, on the other hand, can be achieved in several steps, and there may be applications where no 'true' real-time behaviour is required.

Generally, the degree of complexity of a real-time system will be

determined by the application and will depend upon the required quality of the simulated environment. Possible criteria are, amongst others:

- spatial and temporal stability
- spatial and temporal definition
- naturalness
- authenticity

10.1 Real-time convolution

The binaural room impulse response is an FIR (finite impulse response) filter, and convolution can be performed by determining the convolution product. However, the response length is large and the convolution with an impulse response of 3 s sampled at 48 kHz would require approximately 13 GFlops (floating-point operations) per channel. The required computation time can be drastically reduced to about 11 MFlops/channel for the above example if one uses the well-known overlap-add algorithm.⁴⁴ The disadvantage of this method is that, although all incoming data are processed, a delay of about twice the impulse-response length is induced into the signal flow. This delay will be of the order of 0.5–10 s, depending on the room; this may not be tolerable for some applications. Convolution engines based on this principle are commercially available (e.g. FDP-1, manufactured by Lake, and see Ref. 45).

10.2 Interactivity

One fundamental property of interactivity is that proprioception is supported, i.e. the auditive percepts of the listener are congruent with the corresponding coordinative percepts. For example, if the listener turns his head, the directions of incidence will change. These changes in connection with the head movement are used by the human auditory system to resolve front-back ambiguities and contribute to the spatial stability of an auditory image. A system which is able to offer interactivity is not only able to react in real time on events created by the subject, but also on events related to modifications in the virtual environment itself.

Independently of the type of event, the system becomes time-variant. In the case of continuous events, these variations have to be discretised in time, where the sampling interval has to be chosen with respect to perceptual properties of the human auditory system. The description of the environment in terms of the binaural room impulse response is not valid any more, since the update intervals are small compared to the total length of the impulse response. This problem does not arise if, again, the concept of the

virtual sound sources is used. The impulse responses of the virtual sound sources are rather short and can be considered as being time-invariant within one update interval. Since the contributions of all virtual sources are simply summed at the ear drums of the listener, the overall system remains linear.

The auralisation part of a real-time system for the creation of virtual auditory environment could have the following architecture.⁴⁶ Each virtual source is assigned a pre-filter, holding the information of wall reflections, source directivity, air absorption and propagation attenuation. Propagation delay can be realised by one delay line for all virtual sources. Additionally, a pair of binaural filters is assigned to each virtual source. Alternatively a set of binaural filters with fixed directions can be used, where each virtual source is mapped to the direction which matches the desired direction of incidence best. Another possibility is to split the signal and feed it to the binaural processors of the directions adjacent to the desired ones, where the weighting factors for the binaural processors correspond to the distance between the desired and the fixed directions. This would make use of the psychoacoustical effect of summing localisation.

Such a system deals with different types of events, namely

- rotations of the subject;
- translations of the subject;
- rotations of a sound source;
- translations of a sound source;
- variations in the source's directivity;
- variations in the world's acoustical property;
- variations in the world's geometry.

These events give rise to quite different effects on the auditory event as well as on the required modifications within the system. Rotations of the subject will cause an update of the binaural filters. Rotations of the source and variations of the directivity or the wall properties will result in an update of the pre-filters. The treatment of these events has to be delayed according to the distance of the virtual source with respect to the reflection point to the listener. All other events require a complete re-calculation of the distribution of virtual sound sources. To keep the effects of processing time small, the reaction of the system should be controlled by a priority manager. On the base of a priority list of the properties of the auditory event, this manager will schedule the updates in such a way that the data that are relevant to the most important properties are treated first. For example, to maintain the correct direction of incidence (which corresponds to image stability), the binaural filter of the direct sound has to be updated first according to the

psychoacoustical precedence effect. The priority list can be chosen according to the desired application.

10.3 Reverberation

As already stated in Section 4 the treatment of the reverberation requires special care, even in the case of off-line simulation systems. This aspect is even more important for real-time systems. The number of virtual sources which can be computed and auralised in real-time is obviously limited, and this number will generally be too small to generate reverberation that consists of single reflections. A possible solution to this problem is to use a parametric reverberation algorithm, the parameters of which are derived from the physical data of the room or from the already-computed virtual sources.

In many applications, only translations and rotations of the sound source and the receiver are required. If the reverberant sound field is sufficiently diffuse, its auditory properties do not depend upon the coordinates of source and receiver any more. Pre-computed reverberation, such as an impulse response or as parameters for the corresponding algorithm, may then be used.

11 CONCLUSION

Binaural room simulation is a powerful tool to provide for the auditory component of virtual environment. Since it is completely computer-based, it benefits from the tremendous progress in hardware development, and the method will most likely be available for real-time application before long. The hardware architecture required is fairly straightforward and can easily be interfaced with today's virtual-reality systems. Some equipment, e.g. the head-tracking mechanism, can even be shared.

Several different implementations of binaural room-simulation systems presently exist, with many variable parameters—algorithmic as well as physical ones. There is definitely the need for assessment methods to judge the quality of a simulation with respect to specific criteria. These assessment methods will be used to compare different methods and to optimise the simulation procedures.

However, quite some effort remains to be spent to provide an even deeper insight into the process of human perception in virtual-environment settings. The perception of the auditory environment, like most other perception processes, includes a tremendous amount of information reduction in the transformation from the physical to the perceptive domain,

which is not well understood yet. Research subjects concerning multimodal perception, for example hand-ear coordination or eye-ear coordination, are still in their infancy. There is hope, though, that some parts of the binaural room-simulation process can be further simplified and the quality of the simulation can be improved, taking advantage of improving knowledge of human auditory processing.

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